Previous Year Questions :-

2 marks :-

2018-19 :-

1) Find the value of $i^{3/4}$.

$\Rightarrow$ $i = \cos\frac{\pi}{2} + i\sin\frac{\pi}{2}$

$i^3 = (\cos \pi/2 + i\sin \pi/2)^3$

$= (\cos\frac{3\pi}{2} + i\sin\frac{3\pi}{2})$

$i^{3/4} = (\cos\frac{3\pi}{2} + i\sin\frac{3\pi}{2})^{1/4}$

$= \cos\frac{2k\pi + 3\pi/2}{4} + i\sin\frac{2k\pi + 3\pi/2}{4}$ $[k = 0,1,2,3]$

② Given eqn :- $x^4 + 7x^3 + 2x - 1 = 0$. Apply Descarts' rule.

$\Rightarrow$ $f(x) = x^4 + 7x^3 + 2x - 1$ $\qquad$ $f(-x) = x^4 - 7x^3 - 2x - 1$

$+ \quad + \quad + \quad -$ $\qquad\qquad$ $+ \quad - \quad - \quad -$

number of sign change is 1, so +ve real root :- 1 | number of sign change is 1, so -ve real root :- 1

combination of roots :-

+ve real root	-ve real root	imaginary root
1	1	2

c) $(a+b+c)(bc+ca+ab) > 9abc$

$\Rightarrow$ let a, b, c be three positive quantities :-

Then AM > HM

$$\frac{a+b+c}{3} > \frac{3}{\frac{1}{a}+\frac{1}{b}+\frac{1}{c}}$$

$$\Rightarrow a+b+c > \frac{9}{\frac{bc+ca+ab}{abc}}$$

$\Rightarrow (a+b+c)(bc+ca+ab) > 9abc$ (Proved)

d) Let S be the set of all lines in 3-space. A relation R is defined on S by "$l R m$ iff l lies on the plane of m." for $l, m \in S$. Examine whether R is an equivalence relation.

$\Rightarrow$ Reflexive:- It's reflexive, b/c every line lies on the plane containing itself.

Symmetric:- l lies on the plane of m, but it doesn't mean m will also lie on the plane of l.

So, we won't be checking the Transitivity, b/c it's already determined that R is not an equivalence relation.

e) Let $f: \mathbb{R} \to \mathbb{R}$ is defined by $f(x) = 3x^2 - 5$, find $f^{-1}([70])$

$\Rightarrow$ $3x^2 - 5 = 70$

$\Rightarrow 3x^2 = 75$

$\Rightarrow x^2 = 25$

$\Rightarrow x = \pm 5$

$\therefore f^{-1}([70]) = \{(5, -5)\}$

f) $A = \begin{bmatrix} 3 & -2 \\ 4 & -3 \end{bmatrix}$, find A^{-1}.

$\Rightarrow \therefore |A - \lambda I| = \begin{vmatrix} 3-\lambda & -2 \\ 4 & -3-\lambda \end{vmatrix} = 0$

$\Rightarrow (3-\lambda)\cdot(-3-\lambda) + 8 = 0$

$\Rightarrow \lambda^2 - 9 + 8 = 0$

$\Rightarrow \lambda^2 = 1$

$\Rightarrow \lambda = 1, 1.$

$\therefore$ The ch. matrix :- $\begin{bmatrix} 2 & -2 \\ 4 & -4 \end{bmatrix} \Rightarrow R_2' \to R_2 - 2R_1 \begin{bmatrix} 2 & -2 \\ 0 & 0 \end{bmatrix}$

$R_1' \to \frac{R_1}{2} \begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix}$

The ch. eqn is :- $A^2 - I = 0$

$A^2 = I$

$A^2 \cdot A^{-1} = I \cdot A^{-1}$

$A = A^{-1}$

$\therefore A^{-1} = \begin{bmatrix} 3 & -2 \\ 4 & -3 \end{bmatrix}$

g) Find Dimension:- $S = \{(x, y, w, z) \in \mathbb{R}^4 : x + 2y - z + 2w = 0\}$

$\Rightarrow$ From the given eqn:-

$x + 2y - z + 2w = 0$

$\therefore x = -2y + z - 2w$ ~~$4z = x + 2y + 2w$~~

~~$\therefore S = (x, y, w, z)$~~

~~$= (-2y + z - 2w, y, x + 2y + 2w, z)$~~

~~$= y(-2, 1,$~~

$\therefore S = (x, y, w, z)$

$= (-2y + z - 2w, y, w, z)$

$= y(-2, 1, 0, 0) + w(-2, 0, 1, 0) + z(1, 0, 0, 1)$

So, we can represent it in linear combination with 3 vectors, so the dimension is 3.

h) $T: \mathbb{R}^3 \to \mathbb{R}^3$ defined by $T(x, y, z) = (x + y, y + z, x + 1)$.

$\Rightarrow$ let $u = (x_1, y_1, z_1)$ & $v = (x_2, y_2, z_2)$.

$\therefore T(u) = (x_1 + y_1, y_1 + z_1, z_1 + 1)$

$T(v) = (x_2 + y_2, y_2 + z_2, z_2 + 1)$

$u + v = (x_1, y_1, z_1) + (x_2, y_2 + z_2)$

$= (x_1 + x_2, y_1 + y_2, z_1 + z_2)$

$T(u+v) = ((x_1 + x_2) + (y_1 + y_2), (y_1 + y_2) + (z_1 + z_2), (z_1 + z_2) + 1)$

$\therefore T(u) + T(v) = x_1 + y_1 + x_2 + y_2, y_1 + y_2 + z_1 + z_2; z_1 + z_2 + 2)$

So, $T(u+v) \neq T(u) + T(v)$.

That's why it's not linear.

2019-20

a) If $z+\frac{1}{z}=\sqrt{3}$, then show $z^6=-1$

$\Rightarrow$ $z+\frac{1}{z}=\sqrt{3}$

$\frac{z^2+1}{z}=\sqrt{3}$

$z^2+1=\sqrt{3}z$

$z^2-\sqrt{3}z+1=0$

$\therefore z=\frac{\sqrt{3}\pm\sqrt{3-4.1.1}}{2}$

$=\frac{\sqrt{3}\pm i}{2}$

$=\frac{\sqrt{3}}{2}+i.\frac{1}{2}$ & $\frac{\sqrt{3}}{2}-i.\frac{1}{2}$

$\therefore r=\sqrt{\frac{3}{4}+\frac{1}{4}}=\sqrt{4/4}=\sqrt{1}=1$

$\theta=\tan^{-1}(y/x)=\tan^{-1}\left(\frac{1/2}{\sqrt{3}/2}\right)=\tan^{-1}(1/\sqrt{3})=\pi/6$

$z=r(\cos\pi/6+i\sin\pi/6)$

$z^6=\cos\pi+i\sin\pi$

$=-1$

$\therefore z^6=-1$ (Proved)

b) Show that $n^n\geq n!$ for all $n\in\mathbb{N}$.

$\Rightarrow$ Let $P(n)$ be the statement :- $n^n\geq n!$

$\therefore$ $P(n)$:- $n=1$ $\quad 1^1\geq 1!$

$1\geq 1$

It's true for $n=1$

So, it's also true for $n=k$:- $k^k\geq k!$

now, for $n=k+1$

$\therefore (k+1)^{(k+1)}\geq(k+1)!$

$\Rightarrow (k+1)^k.(k+1)\geq(k+1)\,k!$

$\Rightarrow (k+1)^k\geq k!$

So, it's also true for $(k+1)$. Thus, we can say $n^n\geq n!$ for all $n\in\mathbb{N}$.

c) Give example of relations :- (both are transitive)

i) Symmetric but not reflexive :-

~~$S=\{(1,1),(2,2),(3,3)\}$~~ $S_1=\{(1,2),(2,1),(2,3),(3,2),(1,3),(3,1)\}$

ii) Reflexive but not symmetric :-

~~$S=\{(1,2),(2,1)\}$~~ $S_2=\{(1,1),(2,2),(1,2)\}$

d) Transform the equation to remove the second term:-

$x^3-15x^2-33x+84=0$.

$\Rightarrow$ let $x=y+h$

Given eqn:- $x^3-15x^2-33x+84=0$

if we compare it with $a_0x^n+a_1x^{n-1}+\ldots+a_n=0$

$a_0=1$, $a_1=-15$, $n=3$

To remove the second term:-

$n.a_0h+a_1=0$

$3.1.h+(-15)=0$

$h=5$

~~5~~	~~1~~	~~-15~~	~~-33~~	~~84~~	
	~~0~~	~~-5~~	~~+100~~	~~-335~~	
	~~1~~	~~-20~~	~~67~~	~~-251~~	~~$=A_4$~~
	~~0~~	~~-5~~	~~+125~~		
	~~1~~	~~-25~~	~~192~~		
	~~0~~	~~5~~			
	~~1~~	~~-30~~			

5	1	-15	-33	84	
	0	5	-50	-415	
	1	-10	-83	$-331=A_4$	
	0	5	-25		
	1	-5	$-108=A_3$		
	0	5			
	1	$0=A_2$			
	0				
	$1=A_1$				

So, The required equation is:

$x^3-108x-331=0$

e) $x^4 + 2x^2 + 3x - 1 = 0$

$f(x) = x^4 + 2x^2 + 3x - 1$ $\qquad$ $f(-x) = x^4 + 2x^2 - 3x - 1$

$+ \quad + \quad + \quad -$ $\qquad$ $+ \quad + \quad - \quad -$

No. of +ve ~~real~~ real roots :- 1 $\qquad$ no. of -ve real roots :- 1

So, the possible combination of roots :-

+ve real root	-ve real root	imaginary root
1	1	2

f) $63u + 55v = 1$

$63 = 55 \cdot 1 + 8$

$55 = 8 \cdot 6 + 7$

$8 = 7 \cdot 1 + 1$

$7 = 1 \cdot 7 + 0$

$\therefore \quad 8 - 7 \cdot 1 = 1$

$(63 - 55 \cdot 1) - (55 - 8 \cdot 6) = 1$

$\Rightarrow (63 - 55 \cdot 1) - 55 + (63 - 55) \cdot 6 = 1$

$\Rightarrow 63 - 55 - 55 + 63 \cdot 6 - 55 \cdot 6 = 1$

$\Rightarrow 63(1+6) - 55(1+1+6) = 1$

$\Rightarrow 63 \cdot 7 + 55 \cdot (-8) = 1$

$\therefore u = 7 \quad \& \quad v = -8$

g) Find the dimension of the subspace W of $\mathbb{R}^4$, where:-
$W = \{(x,y,z,w) \in \mathbb{R}^4 : x + 2y - z = 0,\ 2x + y + w = 0\}$

$\Rightarrow$ we can say :- $z = x + 2y \quad \& \quad w = -2x - y$

$\therefore W = \{(x,y,z,w)\}$

$= (x, y, x+2y, -2x-y)$

$= x(1,0,1,-2) + y(0,1,2,-1)$

So, we can represent the subspace in terms of 2 vectors, so the dimension is 2.

h) $T : \mathbb{R}^2 \to \mathbb{R}^2$, $T(x,y) = (2x_1 + y_1, y_1)$, $x, y \in \mathbb{R}^2$

$\Rightarrow$ let α, β be the two vectors.

$\alpha = (x_1, y_1) \quad \& \quad \beta = (x_2, y_2)$

$T(\alpha) = (2x_1 + y_1, y_1)$

$T(\beta) = (2x_2 + y_2, y_2)$

$\alpha + \beta = (x_1 + x_2, y_1 + y_2)$

$T(\alpha+\beta) = (2x_1 + 2x_2 + y_1 + y_2, \ y_1 + y_2)$

$= (2x_1 + y_1, y_1) + (2x_2 + y_2, y_2)$

$= T(\alpha) + T(\beta)$

$\therefore T(\alpha+\beta) = T(\alpha) + T(\beta)$

Now, $T(c\alpha) = \cancel{(2x_1 + y_1, y_1)}$

$= (2cx_1 + cy_1, cy_1)$

$= c(2x_1 + y_1, y_1)$

$= c \cdot T(\alpha)$

$\therefore T(c\alpha) = c \cdot T(\alpha)$

Thus, we can say it's linear.

2021:-

a) Prove that :- $\frac{1}{1\cdot 2}+\frac{1}{2\cdot 3}+\frac{1}{3\cdot 4}+\dots+\frac{1}{n(n+1)}=\frac{n}{n+1}$

⇒ let the statement be $P(m)$.

Now, for $m=1$:-

$$\frac{1}{1\cdot(1+1)}=\frac{1}{1+1}$$

$$\frac{1}{2}=\frac{1}{2}$$

So, it's true for $m=1$.

for $m=k$:-

$$\frac{1}{1\cdot 2}+\frac{1}{2\cdot 3}+\dots+\frac{1}{k(k+1)}=\frac{k}{k+1}$$

for $m=(k+1)$:-

$$\frac{1}{(k+1)(k+2)}$$

∴ L.H.S :- $\frac{1}{1\cdot 2}+\frac{1}{2\cdot 3}+\dots+\frac{1}{k(k+1)}+\frac{1}{(k+1)(k+2)}$

$$=\frac{k}{(k+1)}+\frac{1}{(k+1)(k+2)}$$

$$=\frac{k^2+2k+1}{(k+1)(k+2)}$$

$$=\frac{(k+1)^2}{(k+1)(k+2)}$$

$$=\frac{(k+1)}{(k+2)}=\text{R.H.S}$$ [Prooved]

b) Prove that :- $\frac{(m+1)^m}{2^m}>m!$

⇒ let the quantities be $1,2,3\dots,m$.

Then AM > GM

$$\frac{1+2+3+\dots+m}{m}>\sqrt[m]{1\times 2\times\dots\times m}$$

$$\Rightarrow \frac{\cancel{m}(m+1)}{2}\times\frac{1}{\cancel{m}}>\sqrt[m]{m!}$$

$$\Rightarrow \frac{(m+1)^m}{2^m}>m!$$ (Proved)

c) Show that the product of all values of $(\sqrt{3}+i)^{3/5}=8i$.

$$\Rightarrow \sqrt{3}+i=2\left(\frac{\sqrt{3}}{2}+i\cdot\frac{1}{2}\right)$$

$$=2\left(\cos\frac{\pi}{6}+i\sin\frac{\pi}{6}\right)$$

$$(\sqrt{3}+i)^3=(2)^3\cdot\left(\cos\frac{\pi}{2}+i\sin\frac{\pi}{2}\right)$$

$$(\sqrt{3}+i)^{3/5}=(2^3)^{1/5}\cdot\left(\cos\frac{2k\pi+\pi/2}{5}+i\sin\frac{2k\pi+\pi/2}{5}\right)$$

$$=(8)^{1/5}\left(\cos\frac{2k\pi+\pi/2}{5}+i\sin\frac{2k\pi+\pi/2}{5}\right)\quad [k=0,1,2,3,4]$$

For $k=0$:-

$$8^{1/5}\cdot\left(\cos\frac{\pi/2}{5}+i\sin\frac{\pi/2}{5}\right)$$

For $k=1$:-

$$8^{1/5}\left(\cos\frac{5\pi/2}{5}+i\sin\frac{5\pi/2}{5}\right)$$

for $k=2$:-

$$8^{1/5}\left(\cos\frac{9\pi/2}{5}+i\sin\frac{9\pi/2}{5}\right)$$

For $k=3$:-

$$8^{1/5}\left(\cos\frac{13\pi/2}{5}+i\sin\frac{13\pi/2}{5}\right)$$

for $k=4$:-

$$8^{1/5}\left(\cos\frac{17\pi/2}{5}+i\sin\frac{17\pi/2}{5}\right).$$

Now, multiplying them :-

$$=8^{5/5}\left(\cos\frac{\frac{\pi+5\pi+9\pi+13\pi+17\pi}{2}}{5}+i\sin\frac{\frac{\pi+5\pi+9\pi+13\pi+17\pi}{2}}{5}\right)$$

$$=8\left(\cos\frac{\overset{9}{\cancel{45}}\pi/2}{\cancel{5}}+i\sin\frac{\overset{9}{\cancel{45}}\pi/2}{\cancel{5}}\right)$$

$$=8\cdot\left(\cos 9\,\pi/2+i\sin 9\cdot\pi/2\right)$$

$$=8\cdot\cos\pi/2+i\sin\pi/2$$

$$=8i$$ (Proved)

d) Apply Descart's rule of sign :- $x^4-4x^3+2x^2+4x+1=0$.

$f(x)=x^4-4x^3+2x^2+4x+1$ $\quad$ $f(-x)=x^4+4x^3+2x^2-4x+1$

$+ \; - \; + \; + \; +$ $\quad$ $+ \; + \; + \; - \; +$

No. of sign changes = 2 $\quad$ no. of sign changes = 2

positive real roots = 2 $\quad$ negative real roots = 2

:- Possible combination of roots :-

+ve	-ve	imaginary
2	2	0
2	0	2
0	2	2
0	0	4.

e) What's the smallest equivalence relation on set $A=\{a,b,c\}$

⇒ The smallest equivalence set would be :-
$\{(a,a),(b,b),(c,c)\}$

→ It's reflexive.

→ Symmetric & transitive trivially.

f) $f(x)=|x|+x$. Show it's neither injective nor surjective

⇒ ∴ $f(1)=|1|+1=2$ $\quad$ $f(0)=|0|+0=0$

$f(-1)=|-1|+(-1)=0$

so, it's not injective, because it's producing the same output for two different inputs.
& not surjective because it can't produce negative outputs.

g) $W=\{(x,y,z)\in\mathbb{R}^3 : 2x+y-z=0\}$

⇒ $2x+y-z=0$

$z=2x+y$.

∴ $W=\{(x,y,z)\}=\{(x,y,2x+y)\}$

$=x(1,0,2)+y(0,1,1)$

So, it could be represented as linear combination of 2 vectors, so the dimension is 2.

h) If $A=\begin{bmatrix}1&0&0\\1&0&1\\0&1&0\end{bmatrix}$: find A^{2020}.

→ The cayley-hamilton ch polynomial :- $|A-\lambda I|=0$

∴ $\begin{vmatrix}1-\lambda&0&0\\1&-\lambda&1\\0&1&-\lambda\end{vmatrix}=0$

⇒ $(1-\lambda)(\lambda^2-1)=0$

⇒ $\lambda^3-\lambda^2-\lambda+1=0$

So, A must satisfy this. we can say :- $A^3-A^2-A+I=0$

Here we can see a recurrance relation.

$A^n-A^{n-1}=A^{n-2}-I$

∴ $A^3-A^2=A-I$

$A^4-A^3=A^2-A$

$A^5-A^4=A^3-A^2=A-I$

$- \; - \; -$

$A^{2020}-\cancel{A^{2019}}=A^2-A$

Total equation in the series :- $(2020-3)+1$

$=2017+1$

$=2018$

So, the number of odd & even eq^ns :- $\frac{2018}{2}=1009$

∴ $A^{2020}-A^2=1009(A^2-A)+1009(A-I)$

$=1009A^2-\cancel{1009A}+\cancel{1009A}-1009I$

⇒ $A^{2020}=1010A^2-1009I$

∴ $A^2=\begin{bmatrix}1&0&0\\1&0&1\\0&1&0\end{bmatrix}\times\begin{bmatrix}1&0&0\\1&0&1\\0&1&0\end{bmatrix}=\begin{bmatrix}1+0+0&0+0+0&0+0+0\\1+0+0&0+0+1&0+0+0\\0+1+0&0+0+0&0+1+0\end{bmatrix}$

$=\begin{bmatrix}1&0&0\\1&1&0\\1&0&1\end{bmatrix}$

$$\therefore\ 1010A^2 - 1009I = \begin{bmatrix} 1010 & 0 & 0 \\ 1010 & 1010 & 0 \\ 1010 & 0 & 1010 \end{bmatrix} - \begin{bmatrix} 1009 & 0 & 0 \\ 0 & 1009 & 0 \\ 0 & 0 & 1009 \end{bmatrix}$$

$$A^{2020} = \begin{bmatrix} 1 & 0 & 0 \\ 1010 & 1 & 0 \\ 1010 & 0 & 1 \end{bmatrix}$$

2022:-

a) $z = x + iy$, then prove that $|x| + |y| \leq \sqrt{2}\, z$

$|z| = \sqrt{x^2 + y^2}$.

From cauchy-schwartz inequality :-

$(a_1^2 + a_2^2)(b_1^2 + b_2^2) \geq (a_1 b_1 + a_2 b_2)^2$

$\therefore\ (|x|^2 + |y|^2)\ \cancel{(y^2+1)}\ (1^2 + 1^2) \geq (|x| \cdot 1 + |y| \cdot 1)^2$

$\Rightarrow (|x|^2 + |y|^2) \cdot 2 \geq (|x| + |y|)^2$

$\Rightarrow \sqrt{|x|^2 + |y|^2} \cdot \sqrt{2} \geq |x| + |y|$

$\Rightarrow |x| + |y| \leq \sqrt{2}\,|z|$ (Proved)

b) $2^m > 1 + \sqrt{2^{m-1}}\ m$. Prove this.

$\Rightarrow$ let the quantities be $2^1, 2^2, 2^3, \ldots, 2^{m-1}$

Then AM > GM.

$$\frac{2^1 + 2^2 + 2^3 + \ldots + 2^{m-1}}{m} > \sqrt[m]{2^1 \cdot 2^2 \cdot 2^3 \cdots 2^{m-1}}$$

$$\Rightarrow \frac{2^m - 1}{2 - 1} \cdot \frac{1}{m} > \sqrt[m]{2^{1+2+3+\ldots+m-1}}$$

$$\Rightarrow 2^m - 1 > \sqrt[m]{2^{\frac{(m-1)m}{2}}}\ m$$

2022:-

① $z = x + iy$, Prove $|x| + |y| \leq \sqrt{2}\, z$.

$\Rightarrow z = x + iy$

$|z| = \sqrt{|x|^2 + |y|^2}$

From cauchy-schwartz inequality:-

$(|x|^2 + |y|^2)(1^2 + 1^2) \geq (|x| \cdot 1 + |y| \cdot 1)^2$

$|x|^2 + |y|^2 \cdot 2 \geq (|x| + |y|)^2$

$\sqrt{|x|^2 + |y|^2} \cdot \sqrt{2} \geq |x| + |y|$

$|x| + |y| \leq \sqrt{2}\,|z|$ (Proved)

② $2^m > 1 + \sqrt{2^{m-1}} \cdot m$. Prove this.

$\Rightarrow$ let the quantities be $2^1, 2^2, 2^3, \ldots, 2^{m-1}$

Then AM > GM.

$$\therefore\ \frac{2^1 + 2^2 + \ldots + 2^{m-1}}{m} > \sqrt[m]{2^1 \cdot 2^2 \cdot 2^3 \cdots 2^{m-1}}$$

$$\Rightarrow \frac{2^m - 1}{2 - 1} \cdot \frac{1}{m} > \left(2^{1+2+3+\ldots+m-1}\right)^{1/m}$$

$$\Rightarrow 2^m - 1 > m \cdot 2^{\frac{m(m-1)}{2} \cdot \frac{1}{m}}$$

$$\Rightarrow 2^m > 1 + \sqrt{2^{m-1}} \cdot m \quad \text{(Proved)}$$

③ Prove that a set with n elements has 2^n subsets.

$\Rightarrow$ For $n = 1$:

$S = \{a\}$

we have $2^1 = 2$ sets :- $\phi, \{a\}$.

$n = k$: It's also true $n = k$

$n = k+1$: $S = \{a_1, a_2, a_3, \ldots, a_{k+1}\}$

So, we would get subsets including or excluding a_{k+1}. & the number of subset would be doubled if we do that. So:- $2 \cdot 2^k = 2^{k+1}$

So, it's true for $n = k+1$, hence proved.

④ Find the equation whose roots are the squares of the equation :- $x^4 - x^3 + 2x^2 - x + 1 = 0$.

⇒ let $x = \sqrt{y}$.

$(\sqrt{y})^4 - (\sqrt{y})^3 + 2(\sqrt{y})^2 - \sqrt{y} + 1 = 0$

or, $y^2 - y\sqrt{y} + 2y - \sqrt{y} + 1 = 0$

or, $y^2 + 2y + 1 = \sqrt{y}\,(y+1)$

or, $(y^2+2y+1)^2 = y\,(y+1)^2$

or, $y^4 + 4y^2 + 1 + 4y^3 + 4y + 2y^2 = y(y^2+2y+1)$

or, $y^4 + 4y^2 + 1 + 4y^3 + 4y + 2y^2 = y^3 + 2y^2 + y$

~~or, $y^4 + 4y^3 - y^3 + 4y^2 + 2y + 1 = 0$~~

~~or, $y^4 + 3y^3 + 4y^2 + 2y + 1 = 0$~~

or, $y^4 + 4y^3 - y^3 + 4y^2 + 4y - y + 1 = 0$

or, $y^4 - 3y^3 + 4y^2 + 3y + 1 = 0$

This is the required equation.

⑤ How many symmetric relations can be defined on a set with n elements.

⇒ Each of the n diagonal pairs (a,a) :-
each could be included or not :- 2^n choices.

For non-diagonals a pair (a,b) where $a \neq b$:
either include both (a,b) & (b,a) or exclude both :- $2^{\frac{n(n-1)}{2}}$

So, total :- $2^n \times 2^{\frac{n(n-1)}{2}}$

$= 2^{\frac{n(n+1)}{2}}$ relations can be defined.

~~⑤ How many symmetric elements can be defined on a set with n elements.~~ $x+1-x$ $x+x-2$

~~⇒~~

⑥ Two sets contain 5 & 7 elements respectively. Find the total number of mapping from A to B. How many of them are injective?

⇒ we know that the general formula is :- n^m for m elements to number of mapping from a set with one with n.

& for injective mapping $(m \leq n)$ $P(n,m) = \frac{n!}{(n-m)!}$

So, here :-

total mapping :- $7^5 = 16807$

injective mapping :- $\frac{7!}{(7-5)!} = \frac{7!}{2!} = \frac{7\times6\times5\times4\times3\times\cancel{2}\times\cancel{1}}{\cancel{2}\times\cancel{1}}$

$= 2520$

⑦ A is non-singular matrix of order 4. Find rank.

⇒ As A is a non-singular matrix :-

- $\det(A) \neq 0$
- It has 4 linearly independent rows (or columns)

Now, we know that rank of a matrix is equal with the number of linearly independent rows or columns.

So, rank of $A = 4$.

8) $A = \begin{bmatrix} 1 & 2 & 3 & 1 \\ 2 & 5 & 3 & x \\ 1 & 1 & 6 & x+1 \end{bmatrix} \xrightarrow[R_3' = R_3 - R_1]{R_2' = R_2 - 2R_1} \begin{bmatrix} 1 & 2 & 3 & 1 \\ 0 & 1 & -3 & x-2 \\ 0 & -1 & 3 & x \end{bmatrix}$

$\xrightarrow{R_3' = R_3 + R_2} \begin{bmatrix} 1 & 2 & 3 & 1 \\ 0 & 1 & -3 & x-2 \\ 0 & 0 & 0 & 2x-2 \end{bmatrix}$

For $x = 1$, the rank of the matrix would be 2.

2022-23/GIP-3

① Prove that for any complex number z, $|z| \geq \frac{1}{\sqrt{2}}(|Re\,z| + |Im\,z|)$

$\Rightarrow$ $z = x + iy$

$|z| = \sqrt{x^2 + y^2}$

From Cauchy-Schwartz inequality:-

$$(|x|^2 + |y|^2)(1^2 + 1^2) \geq (|x| \cdot 1 + |y| \cdot 1)^2$$

or, $2 \cdot (|x|^2 + |y|^2) \geq (|x| + |y|)^2$

or, $\sqrt{2} \cdot \sqrt{|x|^2 + |y|^2} \geq |x| + |y|$

or, $|z| \geq \frac{1}{\sqrt{2}}(Re|z| + Im|z|)$ (Proved)

② Construct an equivalence relation on the set $A = \{1, 2, 3\}$

$\Rightarrow$ Smallest equivalence relation on $A = 1, 2, 3$ is

$R = \{(1,1), (2,2), (3,3)\}$

• Justification:-

i) Reflexive :- Every element is related to itself.

ii) Symmetric :- If $(a,a) \in R$ then trivially $(a,a) \in R$ again.

iii) Transitive :- If $(a,a) \in R$ & $(a,a) \in R$, then $(a,a) \in R$.

③ Using principle of mathematical induction prove $3^{2n} - 8n - 1$ is divisible by 64.

$\Rightarrow$ let $P(n) = 3^{2n} - 8n - 1$

for $n = 1$: $3^{2 \cdot 1} - 8 \cdot 1 - 1 = 9 - 8 - 1 = 0$

0 is divisible by 64, so it's true for $n = 1$

for $n = k$: $3^{2k} - 8k - 1 = 64m$

where m is some integer.

for $n = k+1$: $3^{2(k+1)} - 8(k+1) - 1$

$= 3^{2k+2} - 8(k+1) - 1$

$= 3^2 \cdot 3^{2k} - 8k - 1$

$= 9(64m + 8k + 1) - 8k - 1$

$= 9 \cdot 64m + 9 \cdot 8k + 9 - 8k - 1$

$= 9 \cdot 64m + 9 \cdot 8k + 8 - 8k$

$= 3^2 \cdot 3^{2k} - 8k - 8 - 1$

$= 3^2 (64m + 8k + 1) - 8k - 9$

$= 9 \cdot 64m + 9 \cdot 8k + \cancel{9} - 8k - \cancel{9}$

$= 9 \cdot 64m + 64k$

$= 64(9m + k)$

Since $9m + k$ is an integer, $P(k+1)$ is also divisible by 64.

④ Apply Descartes rule:- $x^3 + x^3 - x^3 = 0$

$\Rightarrow$ $x^3 + \cancel{x^3} - \cancel{x^3} = 0 \Rightarrow x^3 = 0$

$f(x) = x^3$ | no sign changes, so 0 positive real roots.

$f(-x) = -x^3$ | no sign changes, so 0 negative real roots.

so, this equation has $x = 0$ with multiplicity 3

⑤ Let λ be an eigenvalue of an $n \times n$ matrix A. Show that λ^4 is an eigenvalue of the matrix A^4.

$\Rightarrow$ Let's suppose :- • A is a $n \times n$ matrix.

• λ is an eigenvalue of A, so there exists a non-zero vector v such that:-

$Av = \lambda v$

$\Rightarrow Av \cdot A^3 = \lambda v \cdot A^3$

$\Rightarrow A^4 v = \lambda \cdot A^2 \cdot (Av)$

$= \lambda \cdot A^2 \cdot (\lambda v)$

$= \lambda^2 \cdot A \cdot (Av)$

$= \lambda^3 \cdot (Av)$

$= \lambda^4 v$

$\Rightarrow A^4 v = \lambda^4 v$

Hence it's proved that matrix A^4 has eigenvalue λ^4.

⑥ Show that the mapping $f(x) = \frac{x}{1-|x|}$, where $S=(-1,1)$ is bijective.

⑦ Prove for any non empty set S, prove that $S \times S$ is an equivalence in S. (2023-24 / GE-3)

⇒ let $S = \{a, b, c\}$

∴ $S \times S$ = all the combination of $(a, b, c) = R$.

$R = \{(a,a), (b,b), (c,c), (a,b), (b,a), (b,c), (c,b), (a,c), (c,a)\}$

In this relation we can see that :-

i) $(a,a), (b,b), (c,c)$ is reflexive & belongs to R & $(a,b,c) \in S$.

ii) $(a,b), (b,a) \in R$ & $(a,b,c) \in S$, where it is symmetric.

iii) $(a,b), (b,c), (a,c) \in R$: $(a,b,c) \in S$.

so it's also transitive.

⑧ Express the matrix A. as a product of elementary matrices, where $A = \begin{bmatrix} 3 & 2 \\ 2 & 5 \end{bmatrix}$.

2022-23 / GE-2

① find the number of positive & negative roots of the equation :- $x^5 - x^4 + x^3 + 8x^2 + 2x - 2 = 0$

⇒ $f(x) = x^5 - x^4 + x^3 + 8x^2 + 2x - 2$

+ − + + + −

no of sign changes is 3, so 3 positive real roots.

$f(-x) = -x^5 - x^4 - x^3 + 8x^2 - 2x - 2$

− − − + − −

no. of sign changes is 2, so 2 negative real roots.

② z is a variable complex number such that an amplitude of $\frac{z-i}{z+1}$ is $\pi/4$. Show that the point z lies on a circle in the complex plane.

⇒ $z = x + iy$

$z - i = x + iy - i = x + i(y-1)$

$z + 1 = x + iy + 1 = (x+1) + y$

Now, argument of $(z-i) = \tan^{-1}\left(\frac{y-1}{x}\right)$

argument of $(z+1) = \tan^{-1}\left(\frac{y}{x+1}\right)$

Argument of $\left(\frac{z-i}{z+1}\right)$ = argument of $(z-i)$ − argument of $(z+1)$ $= \pi/4$

∴ $\tan^{-1}\left(\frac{y-1}{x}\right) - \tan^{-1}\left(\frac{y}{x+1}\right) = \pi/4$

$\tan\left(\tan^{-1}\left(\frac{y-1}{x}\right) - \tan^{-1}\left(\frac{y}{x+1}\right)\right) = \tan \pi/4$

or, $\dfrac{\frac{y-1}{x} - \frac{y}{x+1}}{1 + \frac{y-1}{x} \cdot \frac{y}{x+1}} = 1$

or, $\dfrac{\frac{(y-1)(x+1) - yx}{\cancel{x \cdot (x+1)}}}{\frac{x \cdot (x+1) + (y-1) \cdot y}{\cancel{x \cdot (x+1)}}} = 1$

or, $\dfrac{\cancel{yx} + y - x - 1 - \cancel{yx}}{x^2 + x + y^2 - y} = 1$

or, $y - x - 1 = x^2 + x + y^2 - y$

or, $(x^2 + 2x + 1) + (y^2 - 2y + 1) - 1 = 1$

or, $(x+1)^2 + (y-1)^2 = 1$

This is the formula for circle, where the center point is 1, −1 & radius is 1.

③ Show that for any three real numbers a, b, c :-
$a^8+b^8+c^8 \geq a^2b^2c^2(ab+bc+ca)$

⇒ let the three quantities be a^4, b^4, c^4 :

we will use AM ≥ GM

we know that $x^2+y^2+z^2 \geq xy+yz+zx$

∴ $a^8+b^8+c^8 \geq a^4b^4+b^4c^4+c^4a^4$.

Now, let $x=a^2b^2$, $y=b^2c^2$, $z=c^2a^2$

∴ $a^4b^4+b^4c^4+c^4a^4 \geq (a^2b^2)(b^2c^2)+(b^2c^2)(c^2a^2)+(c^2a^2)\cdot(a^2b^2)$

$\geq a^2b^2c^2(b^2+c^2+a^2)$

Now, $x=a$, $y=b$, $z=c$

∴ $a^2+b^2+c^2 \geq ab+bc+ca$

$(a^2+b^2+c^2)\cdot a^2b^2c^2 \geq c^2b^2c^2(ab+bc+ca)$

So, we can say that $a^8+b^8+c^8 \geq a^2b^2c^2(ab+bc+ca)$

[Proved]

④ Let W be the subspace of $\mathbb{R}^3$ defined by :-
$W=\{(x,y,z)\in \mathbb{R}^3 : 2x+2y+z=0,\ 3x+3y-z=0,\ x+y-3z=0\}$
Find dimension of W.

⇒ $\begin{cases} 2x+2y+z=0 \\ 3x+3y-z=0 \\ x+y-3z=0 \end{cases}$

The co-efficient matrix is :- $\begin{bmatrix} 2 & 2 & 1 \\ 3 & 3 & -1 \\ 1 & 1 & -3 \end{bmatrix}$

Performing elementary row operations :-

$R_1 \rightleftarrows R_3 \begin{bmatrix} 1 & 1 & -3 \\ 3 & 3 & -1 \\ 2 & 2 & 1 \end{bmatrix} \rightarrow \begin{matrix} R_2' = R_2 - 3R_1 \\ R_3' = R_3 - 2R_1 \end{matrix} \begin{bmatrix} 1 & 1 & -3 \\ 0 & 0 & 8 \\ 0 & 0 & 7 \end{bmatrix}$

$R_3' = R_3 - \frac{7}{8}R_2 \begin{bmatrix} 1 & 1 & -3 \\ 0 & 0 & 8 \\ 0 & 0 & 0 \end{bmatrix}$

So, rank of the matrix is 2 & no of unknowns = 3.

∴ dim(W) = no of unknowns − rank of the matrix

$= 3-2$

$= 1$

⑤ If α, β & γ be the roots of the equation $x^3+2x^2-3x-1=0$, Find the value of $1/\alpha\beta + 1/\beta\gamma + 1/\gamma\alpha$.

⇒ $1/\alpha\beta + 1/\beta\gamma + 1/\gamma\alpha$.

From the eqn :-

1) $\alpha+\beta+\gamma = -2$

2) $\alpha\beta+\beta\gamma+\gamma\alpha = -3$

3) $\alpha\beta\gamma = 1$

∴ $1/\alpha\beta + 1/\beta\gamma + 1/\gamma\alpha = \dfrac{\beta\gamma^2\alpha + \alpha^2\beta\gamma + \alpha\beta^2\gamma}{\alpha^2\beta^2\gamma^2}$

$= \dfrac{\alpha\beta\gamma(\alpha+\beta+\gamma)}{(\alpha\beta\gamma)^2}$

$= \dfrac{1\cdot(-2)}{1}$

$= -2$

⑥ The eigen values of the matrix A are 5 & 3. Find the eigen values of the matrix $A^2 - 8A + 15I + 30A^{-1}$

⇒ when $\lambda = 5$:- $(5)^2 - 8\cdot5 + 15 + \overset{6}{\cancel{30}}\cdot\frac{1}{\cancel{5}}$

$= 25 - 40 + 15 + 6$

$= 6$

when $\lambda = 3$:- $(3)^2 - 8\cdot3 + 15 + \overset{10}{\cancel{30}}\cdot\frac{1}{\cancel{3}}$

$= 9 - 24 + 15 + 10$

$= 10$

7) If a is prime to b, prove that a^2 is prime to b & b^2.

$\Rightarrow$ a is prime to b.

so, $\gcd(a,b)=1$

$au+bv=1$ for some integer u,v

$au=1-bv$

$\Rightarrow a^2u^2=1-2bv+b^2v^2$

$\Rightarrow a^2u^2+2bv-b^2v^2=1$

$\Rightarrow a^2u^2+b(2v-bv^2)=1$

$\Rightarrow a^2x+by=1$ [$u^2=x$, $2v-bv^2=y$ are integer]

so, a^2 is prime to b.

$by=1-a^2x$

$\Rightarrow b^2y^2=1-2a^2x^2+a^4x^2$

$\Rightarrow b^2y^2+2a^2x^2-a^4x^2=1$

$\Rightarrow b^2y^2+a^2(2x^2-a^2x^2)=1$ [where $p=y^2$, $2x-a^2x^2=q$ are integers]

so, a^2 is prime to b^2.

2023-24 / GE-3

1) a) Find the values of $(-i)^{2/3}$

$\Rightarrow$ $-i=\cos\frac{\pi}{2}-i\sin\frac{\pi}{2}$ $=\cos\frac{3\pi}{2}+i\sin\frac{3\pi}{2}$ $\cos\frac{2k\pi+3\pi}{5}+i\sin\frac{2k\pi+3\pi}{5}$

$(-i)^2=(\cos\pi/2-i\sin\pi/2)^2=(\cos\pi-i\sin\pi)$

$=\cos(-\pi)+i\sin(-\pi)$

$(-i)^{2/3}=(\cos\pi-i\sin\pi)^{1/3}$

$=(\cos\frac{2k\pi+\pi}{3}-i\sin\frac{2k\pi+\pi}{3})$

b) Apply Descartes' rule of sign: $x^6+4x^4+2x^2+4x+1=0$

$\Rightarrow f(x)=x^6+4x^4+2x^2+4x+1$ | No sign changes, so 0 positive roots.

$f(-x)=x^6+x^4+2x^2-4x+1$ | No. of sign changes $=2$

$+ \quad + \quad + \quad - \quad +$

2 negative real roots.

+ve	-ve	complex
0	2	4
0	0	6

c) construct an equivalence relation on the set $A=\{a,b,c\}$.

$\Rightarrow$ Let S be a set.

$\therefore S=\{(a,a),(b,b),(c,c),(a,b),(b,a),(a,c),(c,a),(b,c),(c,b)\}$

Reflexive:- It's reflexive because of: $(a,a),(b,b),(c,c)\in S$ & $(a,b,c)\in A$

Symmetric:- It's symmetric b/c of :- $(a,b),(b,a),(a,c),(c,a),(b,c)(c,b)\in S$ & $(a,b,c)\in A$

Transitive:- It's transitive b/c of:- $(a,a),(b,b)\rightarrow(a,b)$ & similarly $(a,c),(b,c)\in S$ & $(a,b,c)\in A$.

d) Show that the function $f:\mathbb{R}\rightarrow\mathbb{R}$ defined by $f(x)=|x|+x$ is neither injective nor surjective.

$\Rightarrow$ $f(0)=0+0=0$

$f(-1)=1-1=0$

So, the two diff. roots producing the same output.

$f(-2)=2-2=0$

$f(-3)=3-3=0$

It can't produce the negative real outputs, so it's not surjective.

f) Show that W is a subspace of $\mathbb{R}^4$ defined by: $W=\{(x,y,z,w)\}: x,y,z,w\in\mathbb{R},\ 3x+y+z+2w=0\}$

$\Rightarrow$ $3x+y+z+2w=0$

$z=-3x-y-2w$

$W=\{(x,y,z,w)\}=(x,y,-3x-y-2w,w)$

$=x(1,0,-3,0)+y(0,1,-1,0)+w(0,0,-2,1)$

So, this could be represented with 3 basis vectors, so the dimension is 3.

g) Let λ be an eigen value of an impotent matrix A. Show that λ is 1 or 0.

$\Rightarrow$ For an impotent matrix A, we know: $A^2 = A$.
Let the eigen value's vector be x.

$\therefore \; Ax = \lambda x$

$A \cdot Ax = \lambda x \; A$

$A^2 x = \lambda \cdot A(x)$

$= \lambda \cdot \lambda x$

$= \lambda^2 x$

$\Rightarrow A^2 x = \lambda^2 x$.

Now, again, $A^2 = A$

$\Rightarrow A^2 x = Ax$

$\Rightarrow \lambda^2 x = \lambda x$

$\Rightarrow \lambda^2 x - \lambda x = 0$

$\Rightarrow \lambda(\lambda - 1) \cdot x = 0$ since $x \neq 0$.

$\therefore \lambda = 1$ & $\lambda = 0$

So, we got that $\lambda = 1$ & $\lambda = 0$

h) Find the rank of the matrix A^2, where $A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$.

$\Rightarrow A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$

$\therefore A^2 = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix} \times \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0+0+0 & 0+0+0 & 0+1+0 \\ 0+0+1 & 0+0+0 & 0+0+0 \\ 0+0+0 & 1+0+0 & 0+0+0 \end{bmatrix}$

$= \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$

So, the rank of the matrix = 3

2023-24 / GE-2 :-

① If α, β, γ be the roots of the equation $x^3 + qx + r = 0$, then find the value of $\sum \alpha/\beta$.

$\Rightarrow$ i) $\alpha + \beta + \gamma = 0$

ii) $\alpha\beta + \beta\gamma + \gamma\alpha = q$

iii) $\alpha\beta\gamma = -r$

$\therefore \sum \alpha/\beta = \alpha/\beta + \alpha/\gamma + \beta/\alpha + \beta/\gamma + \gamma/\alpha + \gamma/\beta$.

$= \frac{\beta+\gamma}{\alpha} + \frac{\alpha+\gamma}{\beta} + \frac{\alpha+\beta}{\gamma}$

$= \cancel{\frac{\alpha\beta + \alpha\gamma + \alpha\beta + \beta\gamma + \alpha}{\alpha\beta\gamma}}$

$= \frac{\beta^2\gamma + \beta\gamma^2 + \alpha^2\gamma + \alpha\gamma^2 + \alpha^2\beta + \alpha\beta^2}{\alpha\beta\gamma}$

$= \frac{q \cdot 0 - 3 \cdot (-r)}{-r}$ [From symmetric fns of roots]

$= \frac{3}{-r}$

② Find the value of $(\sqrt{3} - i)^{1/7}$

$\Rightarrow (\sqrt{3} - i) = 2\left(\sqrt{3}/2 - i \cdot 1/2\right)$

$= 2\left(\cos\frac{\pi}{6} - i\sin\pi/6\right)$

$\therefore (\sqrt{3}-i)^{1/7} = 2^{1/7} \cdot \cos\frac{2k\pi + \pi/6}{7} - i\sin\frac{2k\pi + \pi/6}{7}$

③ If, $s = a+b+c$, prove that $\frac{s}{s-a} + \frac{s}{s-b} + \frac{s}{s-c} > \frac{9}{2}$, where a, b, c are unequal positive real numbers.

$\Rightarrow$ Let the 3 quantities be $(s-a), (s-b), (s-c)$

$\therefore$ Their AM > HM

$\frac{s-a+s-b+s-c}{3} > \frac{3}{\frac{1}{s-a} + \frac{1}{s-b} + \frac{1}{s-c}}$

$\Rightarrow \frac{2s}{3} > \frac{3}{\frac{1}{s-a} + \frac{1}{s-b} + \frac{1}{s-c}}$

$\Rightarrow \frac{s}{s-a} + \frac{s}{s-b} + \frac{s}{s-c} > \frac{9}{2}$ (Proved)

d) Show that the mapping $T: \mathbb{R}^2 \to \mathbb{R}^3$ defined by $T(x,y) = e^x, e^y$ is not linear.

$\Rightarrow$ let $\alpha = (x_1, y_1)$ & $\beta = (x_2, y_2)$

$T(\alpha) = (e^{x_1}, e^{y_1})$ & $T(\beta) = (e^{x_2}, e^{y_2})$

Now :- $\alpha + \beta = (x_1 + x_2, y_1 + y_2)$

$T(\alpha+\beta) = (e^{x_1+x_2}, e^{y_1+y_2})$

$= (e^{x_1} \cdot e^{x_2}, e^{y_1} \cdot e^{y_2})$

$T(\alpha) + T(\beta) = (e^{x_1}, e^{y_1}) + (e^{x_2}, e^{y_2})$

$= (e^{x_1} + e^{x_2}, e^{y_1} + e^{y_2})$

So, $T(\alpha+\beta) \neq T(\alpha) + T(\beta)$.

Thus it's not linear.

e) Find the rank of the matrix :- $\begin{bmatrix} 1 & 3 & 5 & -1 \\ 2 & 1 & -2 & 8 \\ 0 & 5 & 12 & -10 \end{bmatrix}$

$\Rightarrow$ $R_2' = R_2 - 2R_1$ $\begin{bmatrix} 1 & 3 & 5 & -1 \\ 0 & -5 & -12 & 10 \\ 0 & 5 & 12 & -10 \end{bmatrix} \to R_3' = R_3 + R_2$ $\begin{bmatrix} 1 & 3 & 5 & -1 \\ 0 & -5 & -12 & 10 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

So, it has 2 non-zero rows, so the rank is 2.

f) Prove :- $a \mid b$ & $a \mid c$ then $a \mid (bx + cy)$.

$\Rightarrow$ $a \mid b \Rightarrow b = am$
$a \mid c \Rightarrow c = an$ [m, n are some integers]

Now, $b = am$

$bx = amx$ —— ①

& $c = an$

$cy = any$ —— ②

① + ② :-

$bx + cy = amx + any$

$= a(mx + ny)$

So, $a \mid (bx + cy)$ [Proved]

g) Let $A = \{1, 2, 3\}$. Verify whether the relation $R = \{(1,1), (2,2), (3,3), (2,3), (3,2), (1,2), (2,1)\}$ is transitive.

$\Rightarrow$ Given $R = \{(1,1), (2,2), (3,3), (2,3), (3,2), (1,2), (2,1)\}$

A relation is called transitive when $(a,a), (b,b), (a,b)$ exists altogether. but here for :- $(1,1)$ & $(3,3)$ there is no $(1,3)$

so the given set is not transitive.

h) If $f: R \to R$ & $g: R \to R$ be two mappings, then compute $f \circ g$ with usual notations :-

$f(x) = x^2 + 3x + 2$ & $g(x) = 3x - 1$ $\forall x \in \mathbb{R}$.

$\Rightarrow$ $\therefore f \circ g = f(g(x)) = x^2 + 3x + 2$

$= (3x-1)^2 + 3(3x-1) + 2$

$= 9x^2 - 6x + 1 + 9x - 3 + 2$

$= 9x^2 + 3x$

$= 3x(3x+1)$